CHAPTER I

INTRODUCTION

In this chapter the researchers will present the introduction of the study such as Situational analysis, Prior arts, Framework of study, Statement of objectives, Time and place of study.

Situation Analysis

A bread spread is a type of spreadable condiment that is typically used on bread, toast or crackers. Bread spread is added to food to enhance the flavor or texture of the food, which may be considered bland without it. Also, bread spread can be sweet or savory. Sweet bread spread includes jams, jellies, and honey. Savory bread spread includes butter, margarine and cream cheese.

However, bread spread can be a good source of essential nutrients such as protein, healthy fats, vitamins and minerals. In recent years, there has been a trend toward healthier bread spread such as nut butters or avocado spread. These spreads are typically high in protein and healthy fats, and they can be a nutritious addition to a balanced diet.

Moreover, bread spread has a long and varied history, and its origins can be traced back to ancient times. In ancient Egypt, people used a mixture of honey and crush nuts as bread spread. In ancient Rome, people would spread a mixture of cheese, herbs and olive oil on bread. In the Middle Ages, bread spread became more widespread as people began to explore new spices and ingredients. Over time, bread spread has evolved and diversified and today there are countless varieties of spread available.

On the contrary, Mung beans (*Vigna radiata*) are small, green beans that belong to the legume family. They have been cultivated since ancient times. While native to India, Mung beans later spread to China and various parts of Southeast Asia. These beans have a slightly sweet taste and are sold fresh, as sprouts or as dried beans (Raman, 2018).

Mung Beans are nutritionally balanced. They contain vitamins, minerals and beneficial enzymes, which makes them an excellent part of a healthy diet (Marengo, 2019). In addition, Mung Beans usually found in market, grocery store and even in sari-sari store. However, Mung beans are usually made as a main dish, shake and fillings which are commonly known to the consumer.

Also, if the Mung beans will take so long to consume, it will affect the texture and the taste of the Mung. And if that happened, there's a possibility that it will not be consumable and marketable. Therefore, the researchers aim to make a bread spread made from Mung Beans which is not common and since it is affordable and accessible.

Prior Arts

There are three prior art that relates to the study of Mug Beans spread that will prove its existence and significance in the society. The prior arts are RU2604936C1, WO2007088540A2, WO2019164651A1.

Prior Art 1. Legume Paste published as RU2604936C1 invented by KHOROSHEVA TATJANA VASILEVNA [RU]; and MORDVINTSEVA ELLADA JUREVNA [RU] and was published on December 20, 2016. FIELD: food industry.SUBSTANCE: invention relates to food industry. Legume paste contains following components, wt%: legumes 50-60, vegetable oil 15-20, biomass of live bifidus and lactic acid bacilli (1:1) 3-5, garlic 3-5, water, salt, spices - balance.EFFECT: proposed pate enables to produce a natural high-quality protein product, correcting microbiocenosis, for dietary nutrition of patients suffering from gastrointestinal diseases and dysbiosis disorders in children and adults.1 cl, 3 ex

Prior Art 2. Concentrate Paste for Dips/ Spread and Process for their Preparation published as WO2007088540A2 invented by BUDMAN ELI HUGO [IL]; and ADATO MARCELO ADRIAN [IL] and was published on August 8, 2007. An edible paste concentrate consisting of ground chickpeas, cooked and dried, and crushed sesame seeds is provided. The paste has a shelf life of at least three months, and when mixed with water, provides a dip or a spread with the flavor and texture of freshly prepared hummus. Also provided is a process for manufacturing a spread or a dip from chickpeas or other legumes.

Prior Art 3. Edible Product Comprising Plant Oils or Creams and Cooked Legumes published as WO2019164651A1 invented by ALTMAN AIDAN [US] and MCCLURE ANDREW [US] and was published on August 29, 2019. The teachings envision a formulation including two or more edible plant oils or creams; and a product resulting from cooking one or more types of legumes. The formulation may be a condiment spread, topping, or cooking or baking ingredient, such as butter. The formulation may further include an edible oil, cream, or milk derived from a plant and having lower fatty acid content than another edible plant oil or cream within the formulation. The formulation may include one or more emulsifying agents, one or more flavoring agents, one or more additives, or a combination thereof. The present teachings also include a process for making the formulation. The teachings relate to a formulation of and method of making an edible product, and more particularly to a formulation of and method of making a creamy, spreadable condiment, topping, or product for use in cooking or baking

As a result, the researchers behind this study invent a new product: Mung Bean Bread Spread. In addition, a homemade spread like Mung Beans Bread Spread is an excellent substitute.

Table 1. Matrix of Comparison

Features	Prior Art 1	Prior Art 2	Prior Art 3	Mung Beans Bread Spread
Smashing	x	x	x	✓
Boiling	✓	x		✓
Blending	x	✓	✓	✓
Sugar	x	x	x	✓
Margarine	x	x	x	✓
Cooking	✓	✓	✓	✓
Rice flour	x	x	x	✓
Knox Gelatin	x	x	x	✓
Cocoa Powder	x	x	x	✓

Framework of the Study

The theoretical framework supported by theories and a conceptual framework using Input-Process-Output, or IPO, was covered in this section by the researchers.

Theoretical Framework

This section provides theories as the basis of the study that are used to the process of making Mung Beans Bread Spread. The study relates to the Heat Transfer Theory. Heat Transfer Theory used in this study, because heat transfer is an operation that occurs repeatedly in the food industry. Whether it is called cooking, baking, drying, sterilizing or freezing, heat transfer is part of the processing of almost every food. An understanding of the principles that govern heat transfer is essential to an understanding of food processing (Boltzmann, 1884). This theory applied to the study because it used heat transfer during the process of boiling mung beans spread to produce the new version of spread.

Additionally, the Theory of blending was included in the study to blend together the liquid mixture and mashed monggo beans. This process quickly blends the mixture smoothly and finely. Since the analysis of observing blender users formed the basis for the development of a new model for blender use, delivering benefits that encourage consumers to rediscover the blender as a useful food preparation instrument (Oppenheimer, A. & Reavy, H., 2003). Therefore, the researchers came up an idea to used blender as another way in preparing a food and since the blender is the only equipment available.

Conceptual Framework

On the other hand, the conceptual framework of the study follows the input, process and output model developed by Asplin (2003). It is a frequently used approach for characterizing the structural process that comprises input, process, and output for conceptualizing the study. The input displayed ingredients, tools and equipment, sources such as review related literature, prior arts and money, all of which contributed significantly to the study.

The process displayed the product development, product testing and evaluation. It also demonstrates the steps and phase measurements as well as an assessment of the product's sensory appearance of the Mung Beans Bread spread. If the product passes the test, it is then evaluated appropriately in terms of smell, appearance, and texture.

In order to develop and improve the product, the researcher solicits thoughts and comments from the evaluator during the assessment phase. Mung beans bread spread is also what is anticipated.

INPUT

Inputs are the materials and resources that the researchers prepared during in making the product to make the expected outcomes. In this study the input are the raw materials in making the Mugs bean spread. The ingredients are Mung beans, Brown Sugar, Cocoa powder, Knox gelatin powder, margarine, salt & vanilla extract. The researchers follow the safety standard in preparing the materials and ingredients to assure its cleanliness.

PROCESS

In this section, it is referred to the process in making product converting input to output. It includes how the materials and tools prepared to make the product. Upon creating the product, it compasses of two processes such as boiling and blending.

OUTPUT

This is the final product that is made from new materials that are processed to come up with the intended output or finish product that is the Mung Beans Bread Spread.

MONGGO BEANS BREAD SPREAD

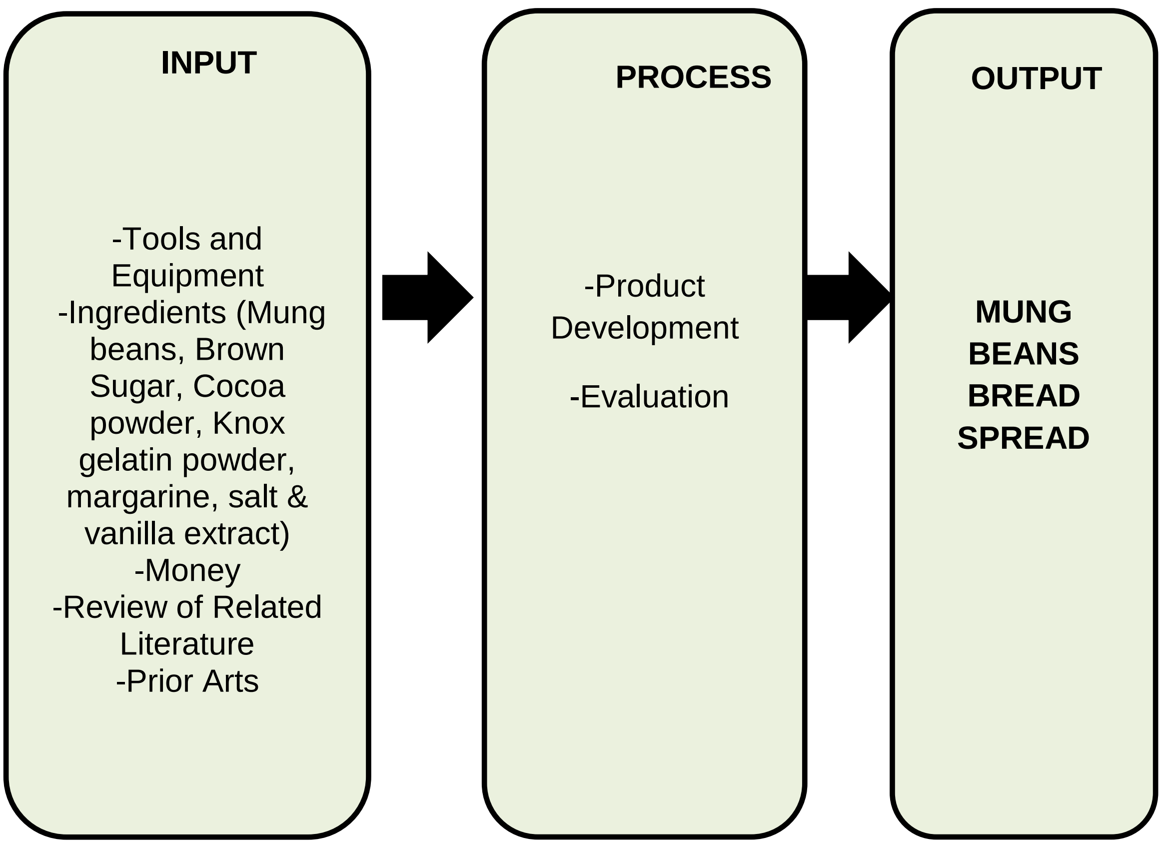

Figure 1. Conceptual Framework of the Study

Statement of the Problem

The main objective of the study is to create a novel and innovative product which is locally available ingredients that produced for the daily consumption of consumers and an opportunity opener for entrepreneurs.

Specific Objectives: To attain the general objective of the study, the following would be sought:

1. To determine the Mung Beans Bread spread process.
2. To present review and related literature on Mung Beans Bread Spread.
3. To determine the satisfaction index on the sensory quality of the product; namely: appearance, aroma, and texture.

Time and Place of the Study

This study entitled “Mung Beans Bread Spread” primarily focuses on processing and producing of spread that conducted in Mati City, Davao Oriental year 2023. This study starts on November 2023 and continued until all the necessary data gathered. In making the mungs beans bread spread it take one day including buying the ingredients that are needed.

This research study will take in to two places such as Mapantad Purok Daisy 1, Barangay Sainz and Purok Pag-asa, Phase 8, Martinez, Mati City. The scopes of this study are limited to sixty-five (65) respondents. The sixty-five respondents are comprised by fifty (50) consumers and fifteen (15) NC II holders.

Operational of Terms

In this section, for a better understanding of this research report, the following terms are operationally defined:

Blending is to combine the mixture or the ingredients from the mashed Mung Beans.

Boiling is the process of heating the water and bringing it to boil the Mung Beans in it in order to cook.

Draining it refers to the process of removing the liquid from the boiled Mung Beans.

Measuring is to get the accurate measurement of ingredients.

Mung Beans it is the main ingredients used in making a product.

Stirring it refers to the process of stirring constantly the mixture during cooking.

Texture it refers to the sense by touch of mouth whether the Mung Beans Bread spread texture smooth, sticky and watery

Mixing refers to the process of combining ingredients together to produce mung beans spread.

CHAPTER II

METHODOLOGY

This chapter presents the tools and procedures in gathering the needed data.

The researcher emphasizes the methodology of the study including the

Research Design, Sources of Data, Materials and Procedures, Instrumentation and Data Collection and Analysis of the Data

Research Design

This study utilizes an experimental-developmental research that employs a quantitative approach. The systematic study of creating, producing, and assessing instructional programs, processes, and products that must meet criteria of internal consistency and effectiveness has been classified as developmental research, (Seels & Richey, 1994).

Consequently, the experimental side took place when we undergone the process of experimenting where the product is applicable and suitable to. The researchers of the study identified appropriate spread product created utilizing the Mung beans. Moreover, pre-experimental research design was applied in this study because there are trials and errors in experimenting the product and it has no control group. For trial the proportion of the ingredients manipulated experiment until to yield desired output.

The ingredients used are; 1 ½ cup mung beans, 5 tbsp. brown sugar, 1 sachet dairy milk and goya chocolates, ½ cup water, and 1/8 tsp. vanilla extract. In the second trial few of the ingredients was changed; 1 1/2 cup boiled mung beans, 5 tbsp. brown sugar, ½ cup water, ½ tsp. cornstarch, 1 tbsp. cocoa powder, 1/8 tsp. vanilla extract and salt to taste. However, the third trial was also changed with 1/3 cup pulverized mung beans, 3 cups water, 2 tbsp. cocoa powder, 6 tbsp. brown sugar, ½ tsp. knox gelatin powder, 1/8 tsp. vanilla extract and salt to taste.

On the other hand, Developmental research has been defined as the systematic study of designing, developing, and evaluating instructional programs, processes, and products that must meet criteria of internal consistency and effectiveness. Developmental research is particularly important in the field of instructional technology. The most common types of developmental research involve situations in which the product-development process is analyzed and described, and the final product is evaluated (Richey, 1994).

Developmental research applied in this study, since the researcher come up with a new variation of spread that will add value to the existing product and creating a new process in making the spread and since it included the process and the product. Moreover, the product is evaluated based on smell, appearance and texture.

Furthermore, the study used purposive sampling technique in choosing the respondent in this study. According to Alchemer, (2021), stated that purposive also known as judgmental, selective, or subjective sampling, where respondents are chosen base on specific criterion. The researchers altered and went beyond the traditional means in their experiment in making homemade Mung Beans Bread Spread.

The main characteristics of a descriptive approach implicate that the researcher uses tools such as questionnaires in order to collect quality of data, foresee future results, or inspect internal relationships as well.

Sources of Data

This section presents the population of the study where to conduct a utilized survey questionnaire as a research method to gather valid data regarding the appearance, aroma, and texture of the Mung Beans which the main ingredient of the product. There will be sixty-five (65) respondents purposively chosen which are comprised by fifty (50) consumer and fifteen (15) NC II holders. The fifty (50) selected consumers must eighteen (18) years old and above since they can provide their impressions and reliable insights regarding the statements that are describing the product. The fifteen (15) selected respondents are the NC II holders in Cookery.

However, before conducting the study the researchers inform the respondents about their rights and responsibilities as part of this study. Moreover, the respondents signed the consent letter given by the researchers before evaluating to prove that they are voluntarily to participate. Nevertheless, upon the respondents evaluating the product the researchers need to read and explain the questions in every criterion. After the evaluation the researcher need to provide groceries worth of two-hundred (200) pesos as a token for their time and participation during conducting of the study.

On the other hand, the respondents are not selected according to their financial status as long as they are willing to participate. The researchers not divulge the research respondent's personal information or status. They are labeled as respondent 1, 2, and 3 and so on. To measure and evaluated the responses of the respondents of this study, the researchers used satisfaction index to evaluate the data regarding the appearance, aroma, and texture. The survey questionnaire indicates feedback, and responses of chosen respondent in terms of appearance, aroma, and texture of Mung Beans Bread Spread product.

Profile of the Product

The food in this study, "Mung Beans Bread Spread," is innovative and has a balanced sweet taste. It also produced another spreads that are novel to their palate.

Materials and Procedures

In this phase, tools and equipment, supplies and materials, and production process of the product presented.

Table 2. Tools, Equipment and Functions

TOOLS AND EQUIPMENT	FUNCTIONS
Casserole	It is where the Mung Beans will be put for boiling purposes.
Stove	It is used for boiling the Mung Beans by providing heat directly.
Spoon	Use to smash the Mung Beans.
Measuring cup	Use to measure the liquid and dry ingredients.
Measuring spoon	Used to measure vanilla extract, cocoa powder, brown sugar and knox gelatin powder
Glass container	Use in storing the finished product.
Strainer	Use for draining the boiled Mung Beans.

	Used to make puree Mung Beans.
Blender	Used to cook the mixture
Pan	It is where the Mung Beans smashed.
Mixing bowl	

Table 3. Materials, Supplies and Costs

MATERIALS AND SUPPLIES	**MARKET PRICE P**	**UNIT AND QUANTITY**	**TOTAL (P)**
Mung Beans	76.00/kg	100g	15.00
Brown Sugar	60.00/kg	1/4g	15.00
Cocoa Powder	72.00/dozen	3sachet	18.00
Knox Gelatin Powder	28.00/each	1sachet	28.00
Vanilla extract	20.00/bottle	5ml	3.00
Salt	10.00/kilo	1pack	3.00
Margarine	75.00/kilo	1pack	15.00
Rice flour	123.00/kilo	1/2kg	63.00
		Total	**320.00**

Table 4. Production Timeframe

Mung Beans Bread Spread Time-Frame of the Production Process	
Nature of Work	Time Allotment
I. Conceptualization of the Product	24 hours
II. Gathering of Tools, Material, and Equipment	2 hours
III. Preparation Process	
a. Mise en Place	30 minutes
b. Sanitation of tools, materials and equipment	30 minutes
c. Boiling	1 hour and 30 minutes
d. Blending	5 minutes
e. Pre-heat	15 minutes
IV. Testing	20 minutes
V. Evaluation	5 hours
Total hour spent	29 hours and 15 minutes

Production Process

Researchers completed the preparatory tasks before producing the final product. The following are included in the preparatory activities: gathering all necessary tools, materials, and equipment; preparing ingredients for the product; and cleaning the area where the mung bean bread spread is made.

The detailed instructions for creating the mung bean bread spread are provided below.

1. Wash and sanitizing the tools needed in making a product.

PLATE NO 1. Wash and sanitize tools

2. Buy 100 grams of Mung beans.

PLATE NO 2. Buying

3. Wash the Mung beans with running water.

PLATE NO 3. Wash Mung Beans

4. Add 4 cups of water.

PLATE NO 4. Adding of water

5. Boil the Mung beans in medium heat until the peel floats.

PLATE NO 5. Boiling

6. When the peel is in the surface, remove it and drain well.

PLATE NO 6. Draining

7. Put boiled Mung beans in a mixing bowl and mash by using spoon.

PLATE NO 7. Smash Mung Beans

8. After, put mashed Mung beans in a blender.

PLATE NO 8. Put in a blender

9. Dissolve 4 tbsp. of cocoa powder, 7 tbsp. of brown sugar, 1/4 tsp. of knox gelatin powder, ½ cup of rice flour and 1 pinch of salt in a ½ cup water.

PLATE NO 9. Mixing

10. Pour the mixture in a blender and blend it together with the mashed Mung beans.


PLATE NO 10. Blending

11. After blending it well, pour it in a low heated pan.

PLATE NO 11. Cooking

12. Stir constantly the Mung beans spread mixture until it becomes sticky and add 1 tsp. (5ml) of vanilla extract and 2tbsp. margarine. Mix it thoroughly.

PLATE NO 12. Add vanilla extract and margarine

13. Set aside the spread and let it cool.

PLATE NO 13. Cooling

14. Store the spread in the refrigerator.

PLATE NO 14. Storing

Revision Phase

This section evaluated the condition and recommendation for revision of the project. The researchers found some defects that were amended right away. The product was measured according to its aroma, texture, appearance. Table 5 showed the revision made from the product.

Table 5. Evaluation of Mung Beans Bread Spread

Critical Parts	Comments/Suggestions	Action Taken
Texture	Add more water to achieve the right consistency	Reduced the amount measurement of cornstarch so that the texture of the product will become watery.
Aroma	Reduce the amount of vinegar	Reduced the amount measurement of vinegar so that the smell not too sour.
Appearance	Reduced the amount of turmeric to have a natural color.	Reduced the amount measurement of vinegar so that the smell not too sour.

Interventions

Table 6 shows the trial and error undergone by the researchers to get the desired taste, texture, aroma and appearance of the product.

Try out	Interventions	Result
Trial 1	The same measurement of salt and sugar.	The taste of the salt dominates
Trial 2	Using cassava starch for thickening.	The result of texture of the product is slimy.
Trial 3	Sugar measurement is only half of salt.	The balance of sweetness and saltiness is good.

Research Instrument and Data Collection

The researcher was used researcher's made survey questionnaire that contains ten (10) items in each criterion with the total of forty (40) items of products description as a research instrument to determine the satiety of the products through survey of satisfaction index of the respondent. Sobrepena (2011) stated that research instrument is a tool used to collect, measure, and analyze data related to your research interests.

The following variable indicators were used to measure the variables involved.

APPEARANCE

SCORE	DESCRIPTIVE RATING	INTERPRETATION
1.00-1.49	Very Unsatisfied (VUS)	The respondents find the product extremely not attractive and dirty.
1.50-2.49	Unsatisfied (US)	The respondents find the product not attractive and dirty.
2.50-3.49	Fair (F)	The respondents find the product undecided on its appearance.
3.50-4.49	Satisfied (S)	The respondents find the product attractive.
4.50-5.00	Very Satisfied (VS)	The respondents find the product very attractive.

AROMA

SCORE	DESCRIPTIVE RATING	INTERPRETATION
1.00-1.49	Very Unsatisfied (VUS)	The respondents find the smell of the product disgusting.
1.50-2.49	Unsatisfied (US)	The respondents find the smell of the having an unpleasant smell.
2.50-3.49	Fair (F)	The respondents find the smell of the product undecided or fairly good.
3.50-4.49	Satisfied (S)	The respondents find the smell of the product aromatic.
4.50-5.00	Very Satisfied (VS)	The respondents find the smell of the product very pleasant in smell.

TEXTURE

SCORE	DESCRIPTIVE RATING	INTERPRETATION
1.00-1.49	Very Unsatisfied (VUS)	The respondents find the product difficult to spread.

1.50-2.49	Unsatisfied (US)	The respondents find the product too thick to spread.
2.50-3.49	Fair (F)	The respondents find the product not totally spreadable.
3.50-4.49	Satisf ied (S)	The respondents find the product spreadable.
4.50-5.00	Very Satisfied (VS)	The respondents find the product spreadable and fine.

Analysis of Data Collection

There are sixty-five (65) respondents in Mapantad Purok Daisy 1, Barangay Sainz and Purok Pag-asa, Phase 8, Martinez, Mati City who were sampled to rate the product in terms of its appearance, aroma, and texture. These number of population composed of 50 consumers and 15 NC II holders in cookery. The data of this study collected through the following procedure:

1. **Seeking permission to conduct the study.** The researchers undergone research ethics such as informing the respondents about their rights and responsibilities as part of this study.
2. **Orientation.** The researcher will inform certain respondents and give a brief explanation about the study for them to understand what the purpose of conducting a study is.

3. **Distribution and Retrieval of the questionnaires.** The researchers personally distribute and retrieve the survey questionnaires to the respondents. The researchers assisted respondents who have difficulties in understanding the item description

4. **Collecting and checking of the questionnaires**. When the respondents done answering, the researcher collect right away and check the survey tool if there are missed items.

Analysis of Data

The data gathered and interpreted using the indicators used in consumer's sensory perception of food attributes. The data interpreted in terms of aroma, texture and appearance. The means used to provide a more detailed description and analysis of the results to find out the significance of the product developed.

There are four (3) criteria such as aroma, appearance, and texture. The item was constructed for every variable and responses rated using the following indicators. The statistical tool that used in this study is mean. It is also known as the average, wherein the total sum of the data is collected (Hurley and Tenny, 2022).

Statistical Treatment

The acquired data was thoroughly evaluated and structured. In order to compute the mean, data was statistically treated using descriptive analysis. Furthermore, the use of an assessment sheet proved the high quality of Mung Beans based on its aroma, texture, and appearance. The significance of this study is determined by the mean statistics.

Ethical Considerations

The primary considerations of this study are the selected respondents who are the custody of the ethics. The researchers ensure the safety of the participants and shoulder everything if there is something incident might happen specifically in evaluating the product and especially when it comes to the health and safety of the participants. The following are some of the rights of research participants.

a. **Informed consent**. The researcher gives all the information to the respondents in terms of the procedures and risk involved in the research. The researchers ensure that the respondents fully understand the research background and they agree that all the information needs to be filled in. Furthermore, the researchers provide a hardcopy of informed consent and give it to the respondents.

b. **Voluntary participation**. The respondents must give free will to participate or to refuse in the said activity. The researchers will not force the respondents to answer the questionnaires. Thus, the respondents have the right to join or not join the said activity.

c. **Confidentiality**. The researchers ensure that all the information, specifically the personal information of the respondents as we practice and protect the privacy act. Thus, the participation of the respondents kept and will not be used without the authority of the respondents.

d. **Anonymity**. The identity of the respondents anonymous and kept for privacy purposes.

e. **Risk of Harm**. The respondents must protected and assured of those possible problems to be faced during conduction of the study. Thus, the researchers shoulder all expenses if there medical medication for the respondents.